

Reducing Food Waste Through Pre-Decision Interaction Design

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Abstract

Household food waste remains a persistent sustainability challenge, even among individuals with strong environmental and health motivations. Prior work in Sustainable Human-Computer Interaction has largely relied on post-decision eco-feedback, which often intervenes too late and can introduce pressure or disengagement in everyday practices. This research investigates an alternative approach: supporting sustainable action before decisions are made, when uncertainty and perceived effort disrupt otherwise strong intentions. Grounded in Self-Determination Theory, the research examines how pre-decision interaction design can support competence without evaluative or guilt-based feedback. Using a research-through-design methodology, FlavorLoop is developed as a competence-supportive design probe that presents a small set of feasible meal options based on ingredients already available at home. A sequential mixed-methods study indicates that pre-decision uncertainty is a key barrier to sustainable cooking, and that constrained, non-evaluative support increases perceived confidence and intention while preserving autonomy.

Keywords

Sustainable HCI, food waste reduction, pre-decision support, intrinsic motivation, Self-Determination Theory, research-through-design, everyday decision-making

ACM Reference Format:

Talayeh Dehghani Ghotbabadi. 2026. Reducing Food Waste Through Pre-Decision Interaction Design. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799170>

1 Introduction

Household food waste remains a persistent sustainability challenge, even among individuals who express strong environmental and health motivations. Across Europe and North America, fresh vegetables are among the most frequently discarded food categories, despite being intentionally purchased for planned meals [1]. This gap between intention and action suggests that food waste is not primarily a problem of awareness or values, but one rooted in breakdowns in everyday decision making. Research in Sustainable Human Computer Interaction has long shown that pro environmental intentions often fail to translate into action when daily

practices are shaped by time pressure, uncertainty, and cognitive fatigue [2, 3]. In domestic contexts, ingredient constraints, unfamiliar produce, and limited mental energy frequently lead to hesitation, avoidance, or default behaviors [4]. To address this gap, many digital interventions rely on post decision eco-feedback, such as energy or carbon metrics, scores, or comparative indicators [5, 6]. While such systems can increase awareness or short term engagement, critical work in Sustainable HCI has demonstrated that outcome focused feedback may also introduce emotional pressure, guilt, or disengagement over time, particularly in routine, low attention practices such as everyday cooking [7–9].

This research starts from a different premise. Sustainability breakdowns often occur before action takes place, at the moment when individuals must decide what to do with the resources already available to them. At this pre decision stage, uncertainty about taste outcomes, perceived effort, or likelihood of success can outweigh even strong sustainability intentions. When confidence is low, postponement or avoidance becomes a rational response, regardless of motivation.

Motivational psychology provides a useful lens for reframing this challenge. Self Determination Theory posits that sustained and self-endorsed behavior depends on the support of autonomy, competence, and relatedness [10, 11]. From this perspective, sustainable cooking fails not because people lack motivation, but because systems often fail to support competence and confidence at the moment of choice. When users feel uncertain, overwhelmed, or implicitly judged, intrinsic motivation becomes fragile and avoidance increases [11, 12].

Motivated by this gap, this student research investigates how pre decision support rather than persuasion or post hoc evaluation can reduce uncertainty and protect intrinsic motivation in everyday sustainable cooking practices. Using a research-through-design approach, FlavorLoop is developed as a design probe to explore emotionally sustainable interaction mechanisms that support confident action at the moment of decision.

2 Related Work

Research in Sustainable Human Computer Interaction has long explored how digital systems can encourage pro-environmental behavior by making environmental impacts visible and actionable. Early work on eco-feedback demonstrated that translating abstract consequences such as energy consumption into concrete, legible metrics can raise awareness and prompt short term behavior change [2, 6]. These approaches have been widely adopted in sustainability oriented systems, particularly in domestic contexts, where feedback, monitoring, and visualization are used to support reflection on everyday practices [3, 9]. However, critical perspectives within Sustainable HCI have questioned the long-term effectiveness and



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ACM ISBN 979-8-4007-2281-3/2026/04

<https://doi.org/10.1145/3772363.3799170>

experiential consequences of outcome focused sustainability interventions. Brynjarsdóttir et al argue that persuasive systems often narrow the framing of sustainability by emphasizing measurable outcomes while neglecting the social, emotional, and contextual dimensions of everyday practice [7]. DiSalvo et al similarly highlight that sustainability design frequently prioritizes optimization and compliance over reflection, care, and lived experience [8]. Research on feedback systems further suggests that post decision evaluation can intervene too late, after decisions have already failed, and may unintentionally introduce pressure or guilt that undermines sustained engagement [9]. Insights from decision making and behavioral psychology help explain these limitations. Studies on choice overload show that large option sets combined with uncertainty about outcomes can reduce motivation and inhibit action, particularly under time pressure [6]. Research on self-control and decision fatigue demonstrates that repeated everyday decisions draw on limited cognitive resources, making it harder to act on long-term intentions when individuals are tired or mentally depleted [9]. The Fogg Behavior Model similarly suggests that behavior occurs when motivation, ability, and prompts converge, and that increasing ability by reducing perceived difficulty may be more effective than increasing motivational pressure [14]. From this perspective, sustainability breakdowns in everyday cooking may reflect insufficient perceived ability at the moment of decision rather than a lack of motivation. These dynamics are especially relevant in domestic cooking, where decisions are frequent, constrained by available ingredients, and often made under low attention.

Motivational theory further clarifies why many sustainability-oriented systems struggle to support sustained behavior change. Self Determination Theory distinguishes intrinsic motivation, which is self-endorsed and resilient, from extrinsic motivation, which depends on external rewards, comparison, or evaluation [10, 11]. Meta analytic evidence shows that extrinsic incentives and performance-based feedback can undermine intrinsic motivation when they introduce pressure or control [12]. In HCI, gamification approaches have frequently operationalized motivation through the use of game design elements such as points, badges, and leaderboards applied to non-game contexts [13], often emphasizing performance metrics and comparison [15, 16], which may conflict with autonomy and emotional safety in everyday practices., which may conflict with autonomy and emotional safety in everyday practices. Recent work therefore calls for emotionally sustainable and care oriented design approaches that support competence and confidence without inducing guilt, anxiety, or performance pressure [8, 17].

Despite this shift, most existing systems continue to focus on tracking, reflection, or evaluation after behavior occurs. Comparatively little attention has been paid to how digital systems can support users at the moment of decision, when uncertainty and hesitation are most acute. This gap motivates the present research, which reframes sustainability support as a pre decision interaction problem rather than a post decision evaluation task.

3 Research Questions and Research Approach

This research examines how digital systems can support sustainable everyday action by intervening before decisions are made,

rather than evaluating behavior after the fact. Building on critiques of outcome-focused sustainability systems, it shifts attention to the moment of decision, when uncertainty, cognitive load, and perceived effort often disrupt otherwise strong sustainability intentions. The central objective is to understand how pre-decision support can reduce uncertainty and protect intrinsic motivation in everyday cooking practices. Rather than persuading users to behave sustainably, this work investigates how interaction design can help individuals act confidently on motivations they already possess, reframing sustainability breakdowns as problems of confidence and competence rather than motivation deficits. Guided by this objective, the research addresses the following questions:

RQ1: How does uncertainty at the moment of decision affect sustainable cooking practices, even among individuals with strong environmental and health motivations?

RQ2: How can pre-decision interaction design support competence and autonomy in everyday cooking without relying on extrinsic rewards, comparison, or guilt-based feedback?

To explore these questions, a research-through-design methodology was adopted, enabling iterative exploration of emerging interaction paradigms through design, reflection, and formative evaluation. Within this approach, FlavorLoop functions as a design probe rather than a finalized or deployable system, allowing the research to focus on decision experience and motivational support rather than system performance or optimization.

A sequential mixed-methods study was conducted across five phases. First, a motivational survey ($N = 117$) examined intentions, perceived barriers, and emotional responses related to sustainable cooking, revealing high intrinsic motivation alongside low confidence at the moment of choice. Second, participatory design workshops ($N = 22$) explored these barriers in depth, surfacing recurring patterns of decision fatigue, fear of wasted effort, and frustration with unfamiliar or surplus ingredients.

Insights from these phases informed the design of FlavorLoop as a pre-decision support probe intentionally constrained to present exactly three feasible meal options based solely on ingredients already available at home. This constraint was designed to reduce choice overload while preserving user autonomy and perceived achievability. Each option is accompanied by a brief, plain-language explanation emphasizing feasibility rather than optimization (e.g., why the dish fits the available ingredients, time, or familiarity level). No scores, rankings, nutritional targets, or sustainability metrics are shown, and options are not ordered or evaluated, allowing the study to focus on how explanations and constrained choice shape confidence and intention rather than performance outcomes.

The probe was evaluated through a storyboard-based study ($N = 50$), expert review ($N = 7$), and formative evaluation ($N = 12$), enabling iterative examination of how pre-decision support influences perceived competence, autonomy, and emotional experience (Table 1). FlavorLoop was implemented as app-like mockups and realistic storyboard scenarios rather than a fully functional system, supporting reflection on interaction mechanisms under realistic everyday constraints.

Table 1: Overview of research phases and objectives

Phase	Method	N	Objective
1	Motivational survey	117	Identify intentions, barriers, and emotional responses related to sustainable cooking
2	Participatory design workshops	22	Explore decision breakdowns and co-design supportive interaction concepts
3	Storyboard-based evaluation	50	Examine perceived confidence and intention under pre-decision support scenarios
4	Expert review	7	Identify design tensions and refine motivational framing
5	Formative evaluation of refined probe	12	Assess autonomy, competence, and perceived achievability

4 Results and Contributions to Date

As this research is ongoing, the results reported here summarize findings to date across five sequential research phases. Rather than evaluating long-term behavioral change, the analysis focuses on how pre-decision support shapes confidence, intention, and motivational experience at the moment of choice, identified as the most critical point of sustainability breakdown.

The motivational survey ($N = 117$) revealed a consistent pattern of high intrinsic motivation toward sustainable eating and food waste reduction alongside persistent difficulty deciding how to use ingredients already available at home. In particular, 78% of participants reported that they “often” or “very often” felt unsure how to use vegetables before they spoiled. Open-ended responses pointed to three recurring sources of uncertainty: anticipated poor taste outcomes, perceived effort or time required, and fear of wasting ingredients if a meal failed. Participants did not frame this hesitation as indifference toward sustainability, but as an inability to act confidently under uncertainty. These findings indicate that sustainability breakdowns frequently occur before action, at the point where individuals must commit to a cooking decision.

Participatory design workshops ($N = 22$) provided qualitative depth to these patterns. Participants described recurring avoidance when faced with unfamiliar, surplus, or seasonal vegetables, often postponing decisions until ingredients spoiled. Rather than expressing disengagement from sustainability values, avoidance was framed as a rational strategy to manage fatigue, perceived risk, and limited time. Participants repeatedly criticized existing digital food tools for overwhelming them with large recipe collections or evaluating outcomes after cooking without supporting decision-making itself. These insights reinforced the importance of intervening at the pre-decision stage rather than after decisions have failed.

To examine how pre-decision support shapes subjective experience, a storyboard-based evaluation was conducted ($N = 50$). Participants compared realistic cooking scenarios with and without pre-decision support. When presented with exactly three concrete and feasible meal options based solely on available ingredients, participants reported higher perceived confidence and stronger intention to use existing ingredients rather than postponing or discarding them. Measures informed by Self-Determination Theory showed consistently high perceived autonomy and competence. These results suggest that narrowing choice sets and clarifying expectations can support intrinsic motivation without relying on extrinsic rewards, scores, or guilt-based feedback.

Expert reviews ($N = 7$) and formative evaluation of the refined design probe ($N = 12$) confirmed the effectiveness of competence-focused pre-decision support while also revealing an important limitation. Support for relatedness remained weak, and passive social cues such as awareness of others’ sustainable behavior did not meaningfully increase motivation. Experts cautioned that increased social visibility could reintroduce comparison, performance pressure, or moral judgment, undermining emotional safety. This finding highlights a central tension in Sustainable HCI, particularly in routine, low-attention practices such as everyday cooking. Table 2 summarizes the key findings across all five research phases, illustrating how pre-decision uncertainty and competence support consistently shaped confidence, intention, and motivational experience.

Taken together, these findings show that pre-decision uncertainty is a key barrier to sustainable cooking even among highly motivated individuals, reframing sustainability breakdowns as problems of confidence and competence, while demonstrating that constrained, explainable choice sets can support intrinsic motivation without evaluative feedback and that emotionally sustainable relatedness remains unresolved for non-comparative social interaction design.

5 Discussion and Future Work

This research reframes sustainable interaction design as supporting confident action under uncertainty rather than persuading users or evaluating behavior after the fact. As illustrated in Figure 1, the shift is not primarily architectural but temporal and experiential: from post-decision evaluative feedback to competence-supportive interaction at the moment of choice. The findings show that many sustainability breakdowns occur at the pre-decision stage of everyday cooking, where uncertainty and anticipated failure outweigh strong sustainability values, challenging the dominance of post-decision eco-feedback in Sustainable HCI. Grounded in Self-Determination Theory, the results indicate that competence support at the moment of choice can enable sustainable action in routine, low-attention practices by reducing uncertainty without relying on guilt, comparison, or performance-based evaluation. While the implications extend beyond cooking to other sustainability-relevant practices, the research also reveals limitations: although pre-decision support increased perceived confidence and intention, support for relatedness remained weak and social visibility risked reintroducing

Table 2: Summary of key findings across research phases

Phase	Key finding
Survey (N=117)	High motivation coexists with low confidence at decision time
Workshops (N=22)	Avoidance framed as rational response to uncertainty
Storyboards (N=50)	Pre-decision support increases confidence and intention
Expert review (N=7)	Competence support effective; social pressure risky
Formative eval. (N=12)	Autonomy preserved; relatedness remains weak

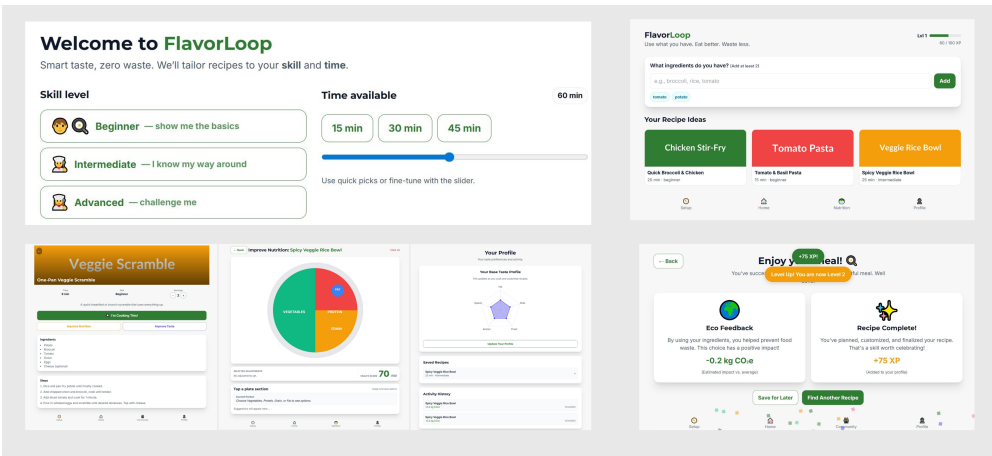


Figure 1: Conceptual contrast between post-decision sustainability feedback and competence-supportive pre-decision interaction, illustrating the shift in *when* and *how* support is provided rather than system architecture.

comparison or pressure. Future work will explore emotionally safe forms of relatedness for pre-decision support.

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